Date: / /

Experiment No. 13

Aim: To study the shift in equilibrium between ferric ions and thiocyanate ions by changing the concentration of either of ions.

Theory:

Ferric chloride when dissolved in water dissociates as,

$$FeCl_3 \longrightarrow Fe^{3+} + 3Cl^-$$

Fe^{3+} ions form a complex in aqueous solution as $[Fe(H_2O)_6]^{3+}$.

Ferric ions react with thiocyanate ion (SCN^-) in aqueous solution, yielding blood-red coloration. A series of complexes are formed by Fe^{3+}-SCN^-- H_2O system by the successive replacement of water molecules from $[Fe(H_2O)_6]^{3+}$ complex by SCN^- ion.

i. $[Fe(H_2O)_6]^{3+} + SCN^- \rightleftharpoons [Fe(H_2O)_5(SCN)]^{2+} + H_2O$

ii. $[Fe(H_2O)_6]^{3+} + 2SCN^- \rightleftharpoons [Fe(H_2O)_4(SCN)_2]^{+} + 2H_2O$

iii. $[Fe(H_2O)_6]^{3+} + 3SCN^- \rightleftharpoons [Fe(H_2O)_3(SCN)_3] + 3H_2O$

These complexes have blood-red coloration. This fact is utilized in the detection of Fe^{3+} basic radical by SCN^- ion in qualitative analysis. The intensity of blood red coloration depends on number of SCN^- ions in the complex.

Apparatus: Burettes, Five 100 mL beakers and test tubes.

Chemicals: $FeCl_3$ (0.01M), NH_4SCN (0.01M) (solutions should be provided to students), distilled water.

Reaction:

$$[Fe(H_2O)_6]^{3+} + 6SCN^- \longrightarrow [Fe(SCN)_6]^{3-} + 6H_2O$$

Procedure:

1. Mix $FeCl_3$ (0.01M), NH_4SCN (0.01M) and distilled water in five different 100 ml beakers in volumes as shown in the following table.

Beaker	0.01M $FeCl_3$ mL	0.01M NH_4SCN mL	Distilled water mL	Intensity of color
I	5.0	15.0	5.0	Lighter
II	5.0	10.0	10.0	Deeper light
III	5.0	5.0	15.0	Blood red
IV	7.5	15.0	2.5	Deeper light
V	2.5	15.0	7.5	Lighter

2. Mix the solution well in every beaker and transfer it to five different test tubes.
3. Compare the colors developed in five different test tubes.

Conclusion:

The balanced chemical reaction suggest that for obtaining better yield of the product $[Fe(SCN)_6]^{3-}$ (blood red coloration), reactant concentration of $FeCl_3$ and NH_4SCN is in ratio 1:3. Therefore the intensity of color in beaker I is the ideal.

Note: 0.01M $FeCl_3$ and 0.01M NH_4SCN.

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MCQ

Select [✓] or underline the most appropriate answer from given alternatives of each sub question.

1. Which color obtained when 0.01M $FeCl_3$ solution is mixed with 0.01M NH_4SCN solution?
   a. Bluish green b. Blood-red c. Violet d. Yellow
2. What is the ratio of reactant concentration of $FeCl_3$ and NH_4SCN to give better yield of the $[Fe(SCN)_6]^{3-}$?
   a. 1:1 b. 1:3 c. 3:1 d. 1:2
3. What is the total volume of 0.01M $FeCl_3$, 0.01M NH_4SCN and distilled water in each beaker?
   a. 10 b. 15 c. 20 d. 25

Short answer questions

1. What is chemical equilibrium?

Ans. Chemical equilibrium is the state of a reversible reaction in which the rate of the forward reaction is equal to the rate of the backward reaction and the concentrations of reactants and products remain constant.

2. Which color obtained, when $[Fe(H_2O)_6]^{3+}$ treated with NH_4SCN?

Ans. A blood-red color is obtained due to the formation of the thiocyanato iron(III) complex.

3. What is the ratio of concentrated NH_4SCN and $[Fe(H_2O)_6]^{3+}$ is taken for better yield of $[Fe(SCN)_6]^{3-}$?

Ans. The ratio is 3 : 1 (NHSCN : [Fe(HO)]³).

4. What is the effect of catalyst on chemical equilibrium?

Ans. A catalyst does not change the position of equilibrium. It only helps the system attain equilibrium faster by increasing the rates of both forward and backward reactions equally.

5. Chemical equilibrium is dynamic in nature. Justify your answer.

Ans. At equilibrium, the forward and backward reactions continue simultaneously at equal rates. Thus, although no net change is observed, the reactions are still occurring, making equilibrium dynamic in nature.

6. Does law of mass action apply on the following reaction?

$$[Fe(H_2O)_6]^{3+} + 3SCN^- \rightarrow [Fe(SCN)_6]^{3-} + 3H_2O$$

Ans. yes The law of mass action applies to this reversible reaction. The equilibrium position depends on the concentrations of the reactants and products.

7. How does concentration of reactant affect direction of chemical equilibrium?

Ans. According to Le Chatelier’s principle, increasing the concentration of a reactant shifts the equilibrium in the forward direction, while decreasing it shifts the equilibrium in the backward direction.

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